

**Exercise 1:** (6 Points) Decide for each of the following statements whether they are true or false. *Hint:* You will get points for correctly answered questions and points subtracted for wrong answers. However, you can not achieve negative points in a given question block.

a) Block I (2 Points)

- i)  $\text{rank}(\mathbf{X}'\mathbf{X}) = p$ , with the number of variables  $p$ .  
☐ TRUE   ☐ FALSE
- ii) Residual plots cannot be used to identify non-linearity in a regression model.  
☐ TRUE   ☐ FALSE
- iii) For the test of significance with  $H_0 : \beta_j = 0$ , it holds that under  $H_0$ ,  $t_j \sim t_{n-p-1}$ .  
☐ TRUE   ☐ FALSE
- iv) In a regression model with a categorical predictor of  $k$  levels,  $k - 1$  dummy variables are created to represent this predictor.  
☐ TRUE   ☐ FALSE

**FFTT**

b) Block II (2 Points)

- i) The presence of multicollinearity can inflate the variance of the coefficient estimates, but does not bias the estimates themselves.  
☐ TRUE   ☐ FALSE
- ii) If the Variance Inflation Factor (VIF) for all variables in a model is close to 1, multicollinearity is likely a concern.  
☐ TRUE   ☐ FALSE
- iii) The coefficient of determination can decrease as more variables are added to the model.  
☐ TRUE   ☐ FALSE
- iv) In the presence of heteroskedasticity, traditional standard errors remain valid for hypothesis testing and constructing confidence intervals.  
☐ TRUE   ☐ FALSE

**TFFF**

c) Block III (2 Points)

- i) The Cook's distance is a measure that helps identifying points that might be unduly influencing the model's fit.  
☐ TRUE   ☐ FALSE
- ii) The "Hat Matrix" in linear regression provides the leverages which help identifying potential outliers.  
☐ TRUE   ☐ FALSE
- iii) AIC and BIC are both criteria for model selection where they penalize models for the number of parameters, but AIC penalizes more heavily than BIC.  
☐ TRUE   ☐ FALSE
- iv) Adjusted  $R^2$  takes into account the number of covariates in the model and can never be negative.  
☐ TRUE   ☐ FALSE

**TTFF**

**Exercise 2:** (6 Points)

Consider the simple linear regression model  $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$  where  $\varepsilon_i \sim N(0, \sigma^2)$ ,  $i = 1, \dots, n$  are independent.

a) Express the model in matrix form, clearly specifying terms  $\mathbf{Y}$ ,  $\mathbf{X}$ ,  $\boldsymbol{\beta}$ , and  $\boldsymbol{\varepsilon}$ . (1 Point)

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \dots \\ Y_n \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \dots & \dots \\ 1 & x_n \end{bmatrix} \cdot \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \dots \\ \varepsilon_n \end{bmatrix}$$

b) Write down the matrix  $\mathbf{X}'\mathbf{X}$ . Under which conditions is the matrix  $\mathbf{X}'\mathbf{X}$  invertible? (2 Points)

$$\mathbf{X}'\mathbf{X} = \begin{bmatrix} 1 & 1 & \dots & 1 \\ x_1 & x_2 & \dots & x_n \end{bmatrix} \cdot \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \dots & \dots \\ 1 & x_n \end{bmatrix} = \begin{bmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{bmatrix} \quad (1pt)$$

Condition:  $\mathbf{X}'\mathbf{X}$  should be full-rank/non-singular, and/or  $\mathbf{X}$  should be full-rank/non-singular matrix. (1pt)

c) Derive the least square estimators in matrix form. (3 Points)

$$\boldsymbol{\varepsilon} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}$$

$$RSS = \boldsymbol{\varepsilon}'\boldsymbol{\varepsilon}$$

$$= (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

$$= \mathbf{y}'\mathbf{y} - \mathbf{y}'\mathbf{X}\boldsymbol{\beta} - \boldsymbol{\beta}'\mathbf{X}'\mathbf{y} + \boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta} \quad (1pt)$$

$$\text{OLS estimation: } \frac{\partial RSS}{\partial \boldsymbol{\beta}} = -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\boldsymbol{\beta} = 0 \quad (1pt)$$

Solved answer:

$$\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{X}^{-1}\mathbf{X}'\mathbf{y} \quad (1pt)$$

**Exercise 3:** (18 Points)**Data set**

This **state** dataset contains US State Facts and Figures. Table 1 includes names and descriptions of the variables. The sample size is  $n = 50$ .

Table 1: Description of the data set

| Variable   | Description                                    | Mean      | Std. dev. | Min/max       |
|------------|------------------------------------------------|-----------|-----------|---------------|
| Population | population estimate as of July 1, 1975         | 4246.000  | 4464.491  | 365/21198     |
| Income     | per capita income (1974)                       | 4436.000  | 614.469   | 3098/6351     |
| Illiteracy | illiteracy (1970, percent of population)       | 1.170     | 0.609     | 0.500/2.800   |
| Life.Exp   | life expectancy in years (1969–71)             | 70.880    | 1.342     | 67.96/73.60   |
| Murder     | murder rate per 100,000 population (1976)      | 7.378     | 3.691     | 1.400/15.100  |
| HS.Grad    | percent high-school graduates (1970)           | 53.110    | 8.076     | 37.80/67.30   |
| Frost      | number of days with temperature below freezing | 104.460   | 51.980    | 0.000/188.000 |
| Area       | land area in square miles                      | 70736.000 | 85327.299 | 1049/566432   |

The following are extracts from the output of fitting and comparing linear regression models in R:

Call:

```
lm(formula = Life.Exp ~ log(Population) + Income + Illiteracy +
    Murder + HS.Grad + Frost + Area, data = state)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-1.44732 -0.46152  0.04912  0.51562  1.63542
```

Coefficients:

|                 | Estimate      | Std. Error   | t value | Pr(> t )   |
|-----------------|---------------|--------------|---------|------------|
| (Intercept)     | 68.3905942620 | 2.1984172195 | 31.109  | 0.0000 *** |
| log(Population) | 0.2707464551  | 0.1347996394 | 2.009   | 0.0510 .   |
| Income          | -0.0000414478 | 0.0002424471 | -0.171  | 0.8651     |
| Illiteracy      | 0.0946136867  | 0.3685371928 | 0.257   | 0.7986     |
| Murder          | -0.2980379797 | 0.0452409716 | (A)     |            |
| HS.Grad         | 0.0583917289  | 0.0242256035 | 2.410   | 0.0204 *   |
| Frost           | -0.0046454782 | 0.0032769627 | -1.418  | 0.1637     |
| Area            | 0.0000002161  | 0.0000016658 | 0.130   | 0.8974     |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: (B) on ?? degrees of freedom

Multiple R-squared: (C), Adjusted R-squared: 0.6981

F-statistic: (D) on ? and ?? DF, p-value:

- a) What are the dimensions (i.e. number of rows and columns) of  $(\mathbf{X}'\mathbf{X})^{-1}$  in this example? (1 Point)  
 Find  $n = 50$  from the question, find  $p = 7$  from the regression table. Then the dimension of the matrix  $\mathbf{X}'\mathbf{X}$  is the same as  $(\mathbf{X}'\mathbf{X})^{-1}$ , which is  $(p+1) \times n \times n \times (p+1)$ , which will be a  $8 \times 8$  matrix.

- b) Reproduce the numerical values replaced in the above output by (A), (B), (C), and (D). Hint: You may find the information that the sum of squared residuals  $\hat{\epsilon}'\hat{\epsilon} = 22.84961$  useful. (4 Points)

$$(A): t = \frac{\hat{\beta}_j}{se(\hat{\beta}_j)} = \frac{-0.298}{0.045} = -6.587$$

$$(B): \hat{\sigma}^2 = \frac{\hat{\epsilon}'\hat{\epsilon}}{n-p-1} = \frac{22.849}{42} = 0.5440, \text{ given this, } \hat{\sigma} = \sqrt{0.5440} = 0.7375.$$

$$(C): R_{adj}^2 = 1 - \left( \frac{1-R^2}{n-p-1} \cdot (n-1) \right) \rightarrow R^2 = 1 - \frac{n-p-1}{n-1} \cdot (1 - R_{adj}^2) = 1 - \frac{42}{49}(1 - 0.6981) = 0.7412$$

$$(D): F = \frac{R^2}{1-R^2} \cdot \frac{n-p-1}{p} = \frac{0.7412}{1-0.7412} \cdot \frac{42}{7} = 17.1864$$

- c) What is the interpretation of the coefficients of the independent variable **Population**? (1 Point)

As population increases by 1%, the average life expectancy will increase by  $0.01 * 0.27 = 0.0027$  units, given other conditions unchanged.

If on average or given other conditions unchanged is missing, -0.5pt.

- d) Based on your calculation in a), conduct a significance test for the independent variable **Murder**. Please clearly write the null and alternative hypotheses, your test statistic, degrees of freedom, critical value, and draw the correct conclusion based on your calculation. You may find the information on page 8 (Table 2-4) useful. Assume  $\alpha = 0.05$ . (2 Points)

Null hypothesis:  $H_0 : \beta_{Murder} = 0$

Alternative hypothesis:  $H_1 : \beta_{Murder} \neq 0$ .

Test distribution: t-distribution

$$\text{t-statistic: } t = \frac{\hat{\beta}_j}{se(\hat{\beta}_j)} = \frac{-0.298}{0.045} = -6.587$$

$$\text{Degrees of freedom: } n - p - 1 = 50 - 7 - 1 = 42$$

Critical value:  $t_{0.975, 42} = 2.0180$  (Here if the student is confused to use other degrees of freedom or other significance level, -0.5)

Conclusion: Since it's a two-sided test, the absolute value of the test statistic is far larger than the critical value. We reject the null hypothesis. The coefficient under a 0.05 significance level are significantly different from 0.

- e) Based on your calculation in a), conduct a significance test for the regression model. Please clearly write the null and alternative hypothesis, your test statistic, degrees of freedom, critical value, and draw the correct conclusion based on your calculation. You may find the information on page 8 (Table 2-4) useful. Assume  $\alpha = 0.05$ . (3 Points)

Null hypothesis:  $H_0 : \beta_1 = \beta_2 = \dots = \beta_7 = 0$

Alternative hypothesis: At least one of the coefficients is not equal to 0.

Test distribution: F-distribution

Test-statistic:  $F = \frac{R^2}{1-R^2} \cdot \frac{n-p-1}{p} = \frac{0.7412}{1-0.7412} \cdot \frac{42}{7} = 17.1864$

Degrees of freedom:  $df_1 = p = 7, df_2 = n - p - 1 = 42$ .

Critical value:  $F_{7,42,0.95} = 2.2371$ . (Here if the student is confused to use other degrees of freedom or other significance level, -0.5)

Conclusion: Since it's a one-sided test, the value of the test statistic is far larger than the critical value. We reject the null hypothesis. At least one of the regression coefficients is not equal to 0.

- f) Calculate the AIC and the BIC of the model on page 3. In general, how do you choose a model using the measures of AIC and/or BIC? And What is the difference between the two measures? (4 Points)

$$AIC = n \cdot \log(RSS/n) + 2 \cdot (p + 1) = 50 \cdot \log(22.84961/50) + 2 \cdot 8 = -23.154$$

$$BIC = n \cdot \log(RSS/n) + \log(n) \cdot (p + 1) = 50 \cdot \log(22.849/50) + 8 \cdot \log(50) = -7.858$$

Here note that in the regression book page 148, the ML estimator of the variance  $\hat{\sigma}^2 = RSS/n$  is considered in AIC and BIC, and not the usual unbiased variance estimator.

Choose criterion: Both measures are used to choose the best model, the smaller the measures, the better the model. However, AIC and BIC cannot be used as direct comparisons, if the student answered as: Choose AIC because AIC is smaller, then it's wrong.

Difference: From a practical point of view, the main difference is that the BIC penalizes complex models much more than the AIC. Thus, the resulting "best" models are typically more parsimonious when using the BIC rather than the AIC.

- g) Calculate the 95%-confidence interval for  $\beta_4$  (HS.Grad). Is your calculated confidence interval consistent with the hypothesis test result given in the regression table? Justify your answer. (2 Points)

Formular:  $CI = \hat{\beta}_j \pm t_{\alpha/2, df} \cdot se(\hat{\beta}_j)$

Results:  $0.0583 \pm 0.0242 \cdot t_{0.975, 42} = 0.0583 \pm 0.0242 \cdot 2.0180 = [0.00950, 0.10727]$ . (1pt)

The result is consistent with the hypothesis test result given in the regression table since 0 is not contained in the confidence interval. In other meaning, we could reject the null hypothesis that the coefficient is 0 under a 0.05 significance level. (1pt)

- h) How do you interpret the  $R^2$  term? Why is there a difference between the  $R^2$  term and the adjusted- $R^2$  term? Justify your answer (1 Points)

$R^2$  is interpreted as how much of the variation of the dependent variable can be explained by the variation of the model. In this case, we could say that about 74.12% variation of life expectancy is explained by the change of the covariates.

Difference between  $R^2$  and adj- $R^2$ : Since as more independent variable is added to the model  $R^2$  will always increase. Therefore the adj- $R^2$  added the penalty of number of covariates. It can only increase if real influential variable is added to the model, it can also be negative if the model is a poor fit.

## Quantile tables

| Table 2: Quantiles for some $t$ -distributions |         |        |        |        |        |        |
|------------------------------------------------|---------|--------|--------|--------|--------|--------|
| quantile                                       | 0.495   | 0.9    | 0.95   | 0.975  | 0.99   | 0.995  |
| 3                                              | -0.0136 | 1.6377 | 2.3534 | 3.1824 | 4.5407 | 5.8409 |
| 7                                              | -0.0130 | 1.4149 | 1.8946 | 2.3646 | 2.9980 | 3.4995 |
| 12                                             | -0.0128 | 1.3562 | 1.7823 | 2.1788 | 2.6810 | 3.0545 |
| 13                                             | -0.0128 | 1.3502 | 1.7709 | 2.1604 | 2.6503 | 3.0123 |
| 25                                             | -0.0126 | 1.3163 | 1.7081 | 2.0595 | 2.4851 | 2.7874 |
| 42                                             | -0.0126 | 1.3020 | 1.6819 | 2.0180 | 2.4184 | 2.6980 |
| 49                                             | -0.0126 | 1.2990 | 1.6766 | 2.0095 | 2.4048 | 2.6799 |
| 50                                             | -0.0126 | 1.2987 | 1.6759 | 2.0086 | 2.4033 | 2.6778 |
| 100                                            | -0.0126 | 1.2901 | 1.6602 | 1.9840 | 2.3642 | 2.6259 |

| Table 3: Quantiles for the standard normal distribution |         |        |        |        |        |        |
|---------------------------------------------------------|---------|--------|--------|--------|--------|--------|
| quantile                                                | 0.495   | 0.9    | 0.95   | 0.975  | 0.99   | 0.995  |
|                                                         | -0.0125 | 1.2816 | 1.6449 | 1.9600 | 2.3263 | 2.5758 |

| Table 4: Quantiles for some $F$ -distributions with fixed $df_2 = 49$ |        |        |        |        |        |        |
|-----------------------------------------------------------------------|--------|--------|--------|--------|--------|--------|
| quantile                                                              | 0.495  | 0.9    | 0.95   | 0.975  | 0.99   | 0.995  |
| $df_1 = 3$                                                            | 0.7905 | 2.1991 | 2.7939 | 3.3961 | 4.2084 | 4.8379 |
| 4                                                                     | 0.8425 | 2.0633 | 2.5611 | 3.0602 | 3.7283 | 4.2430 |
| 5                                                                     | 0.8747 | 1.9685 | 2.4044 | 2.8384 | 3.4162 | 3.8596 |
| 6                                                                     | 0.8965 | 1.8980 | 2.2904 | 2.6793 | 3.1948 | 3.5893 |
| 7                                                                     | 0.9123 | 1.8431 | 2.2032 | 2.5587 | 3.0285 | 3.3871 |
| 8                                                                     | 0.9246 | 1.7989 | 2.1340 | 2.4637 | 2.8983 | 3.2294 |
| 9                                                                     | 0.9336 | 1.7625 | 2.0775 | 2.3866 | 2.7932 | 3.1025 |
| 10                                                                    | 0.9411 | 1.7319 | 2.0303 | 2.3226 | 2.7064 | 2.9979 |

| Table 5: Quantiles for some $F$ -distributions with fixed $df_2 = 42$ |        |        |        |        |        |        |
|-----------------------------------------------------------------------|--------|--------|--------|--------|--------|--------|
| quantile                                                              | 0.495  | 0.9    | 0.95   | 0.975  | 0.99   | 0.995  |
| $df_1 = 3$                                                            | 0.7924 | 2.2191 | 2.8270 | 3.4457 | 4.2853 | 4.9396 |
| 4                                                                     | 0.8444 | 2.0837 | 2.5943 | 3.1089 | 3.8021 | 4.3394 |
| 5                                                                     | 0.8766 | 1.9894 | 2.4377 | 2.8866 | 3.4882 | 3.9528 |
| 6                                                                     | 0.8985 | 1.9193 | 2.3240 | 2.7273 | 3.2658 | 3.6804 |
| 7                                                                     | 0.9143 | 1.8649 | 2.2371 | 2.6068 | 3.0988 | 3.4768 |
| 8                                                                     | 0.9263 | 1.8211 | 2.1681 | 2.5118 | 2.9681 | 3.3181 |
| 9                                                                     | 0.9356 | 1.7850 | 2.1119 | 2.4348 | 2.8628 | 3.1906 |
| 10                                                                    | 0.9432 | 1.7547 | 2.0650 | 2.3710 | 2.7758 | 3.0855 |